

Doman's inclined floor method for early motor organization simulated with a four neurons robot

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Abstract. *Early motor organization using the Domans inclined floor method is simulated with a four neurons robot. A LEGO robot controlled by a biologically plausible neural network performs the same kind of “inclined floor” training that is given by parents to young babies for early motor organization. When the inclined floor training is applied to the robot, it organizes its motor behavior in a manner that is analogous to the motor organization seen in babies. In this way the simulation with the robot could help to understand the kind of neural processes that are involved in early motor organization.*

Keywords: Keywords: Doman's method, inclined floor, neurological re-organization, neural networks, intrinsic plasticity, synaptic plasticity.

1 Introduction

Temple Fay (1895-1963) was a famous neurosurgeon who developed well established surgery techniques like the tentorial section technique for large acoustic neuromas, the fluid restriction technique in head trauma and refrigeration techniques for the control of cancer. Working with physiotherapist Glenn Doman, he led a team of medical professionals who developed several techniques for improving the intellectual and motor skills of normal and brain-injured children. They founded “The Institutes for the Achievement of Human Potential” in Pennsylvania.

Some of the techniques developed by the institutes (see www.iahp.org) appear in many videos that were recorded and placed on the internet by parents all over the world. Regarding intellectual skills, these videos show how less than one-year infants are capable of reading and even performing mathematical operations such as addition and multiplication.

Regarding motor skills, these videos also show how infants start to crawl with less than two months old when they are stimulated through the use of the “inclined floor” technique.

The inclined floor stimulation technique[3] consists of placing a child in a prone position over an inclined floor or ramp, looking to the lower end of the ramp. The ramp should be slippery and inclined in such an angle that when the

child moves forward, he/she slips and falls a little over the ramp. The lateral walls of the ramp are painted in black and white stripes[4]. (Doman et al. 2006) Parents are encouraged to cheerfully make sounds like clapping their hands when the baby shifts forward over the ramp. Glenn Doman describes the foundations of this method:

“When a well infant is put on the floor on a belly-down position he does not know how to crawl... and his arms and legs moves freely... these movements will occasionally result in pushing him forward. When this has happened by accident over and over again, the baby discovers which movements have that effect and which do not. He then discovers how the movements that push him forward feel as distinguished from how the movements that do not push him forward feel. He finally discovers how to purposely reproduce those motions that push him forward and how to synchronize them with each other into a pattern.”[3]

Temple Fay and Glenn Doman believe that the nervous system is a sophisticated cybernetic system in exactly the terms Norbert Wiener defined it in his book *Cybernetics*. [11]

“It was clear that the human brain, like Dr. Wiener's self-regulating system, operated like a cybernetic loop. In human beings this loop, which begins in the environment, follows sensory pathways to the brain and motor pathways from the brain back to the environment.”[3]

It was this idea that stimulated us to construct a simple cybernetic system in the form of a little robot (see Fig. 1) with a rudimentary sensory system that consisted in an acceleration detector (mimicking the vestibular system), a microphone (for mimicking audition) and a light detector sensor (as a rudimentary visual system). The information from this sensory system arrives to a four neurons neural-network that could be seen as the robot's nervous system. Each of these neurons is interconnected to each other and also to two servomotors. The motor pathway of our rudimentary cybernetic system is represented by the outputs of the neurons that command the two servomotors. Each neuron activates either a clock or anti-clock wise direction of each motor.

The environment in our cybernetic system is represented by a ramp, by the black and white stripes painted on the ramp and by the sound produced by a baby shaker.

For describing the cybernetic loop that controls the dynamic of the robot we will paraphrase the above Glenn Doman's explanation:

“When the robot is put on the ramp it is not initially predisposed to move in any specific direction and its wheels responds to its neuron's commands by moving randomly, occasionally resulting in pushing the robot down the ramp. This part of the system, from neurons to wheels, corresponds to the feed forward part of the robot's cybernetic loop. When the robot's descension happened by accident over and over again, the robot ‘discovers’ how the movements that push him forward feel as distinguished from how the movements that do not push him forward feel. The difference between these two kinds of movements is in the size of the sensory inputs because, when descending, the robot slips a little aided by gravity; this fact produces a greater downward acceleration

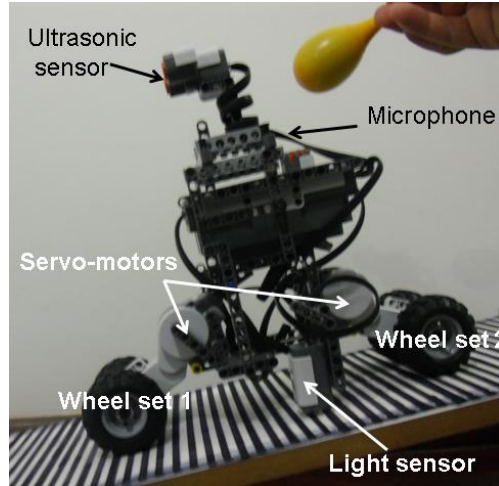


Fig. 1. The Robot

(perceived by the acceleration detector). When the robot descends, the greater downward acceleration also makes the black stripes alternate quicker in front of the light detector than when the robot ascends. Another kind of stimulus that appears when the robot moves downwards is the sound of the baby-shaker. This sound stimulation tries to mimic the encouraging words of parents when the baby performs downward movements over the ramp. This part of the system, from environment to neurons, corresponds to the feedback part of the robot's cybernetic loop.

The robot finally discovers how to purposely reproduce those motions that push him downwards and how to synchronize them with each other into a pattern. This happens, as will be shown, because a successful downward movement of the robot is immediately followed by an intense rewarding stimulation from sensory inputs to the neurons that contributed to the movement, allowing enhanced plasticity processes between these neurons."

A similar kind of process has been detected in real organisms. In the case of the vestibular system, Takeda and collaborators[10] discovered that vestibular stimulation enhances acetylcholine production in the rat hippocampus (a brain structure involved in short-term memory). Related to this, Manabe and collaborators research[7], shows that acetylcholine is a neuro-modulator that contributes to synaptic weights modulation or, in other words, to plasticity processes between neurons.

At the light of the above processes, it would not be very audacious to hypothesize that when a child is experiencing body acceleration (i.e. vestibular stimulation) due to inclined floor therapy, plasticity processes in his hippocampus could help him to consolidate short term memories related to the motor tasks he is performing (as for example learning how to crawl).

This paper is organized as follows. In section two we present the robotic system and the neural architecture that allows the robot to learn. We also explain how neurons were modeled highlighting the cybernetic principles that are implicit in the equations. In section 3 we place the robot on the inclined floor, and evaluate its behavior with different kind of stimuli.

In section 4 we study the practical consequences that could be derived from the tests with the robot for improving the performance of the inclined floor technique.

2 Architecture of the robot

2.1 Architecture and connectivity of the robotic system

The robotic structure was built with a LEGO MindstormsNXT toolkit. The software for modeling the neural network and the sensory and motor commands were written in Matlab using the RWTH Mindstorms NXT toolbox which is of public domain and available on the internet.

The feed-forward path of the robot consisted in a set of four fully interconnected neurons (see Fig. 2), each one of them controlling a specific clockwise or anticlockwise movement of two servomotors acting over two wheel sets. The feed-back path of the robot consisted in three sensors: a microphone, a light sensor and an ultrasonic sensor. The contribution of these sensors is summed and this sum increases the activation of the four neurons.

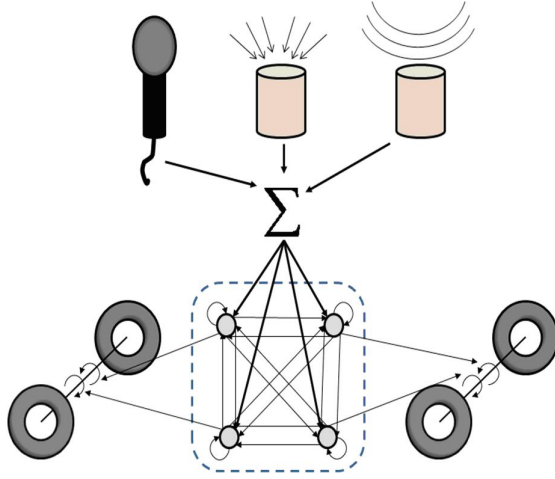


Fig. 2. Organization of the robotic system. Three types of sensors send their information to the four-neurons robot “brain”. Each neuron commands a specific clock or anticlockwise movement of each of the two wheel sets.

2.2 Neural organization of the robot

As previously mentioned, the robot is controlled by the activity of a four-neurons' neural-network. This fully interconnected network is composed of four biologically plausible excitatory-rate code neurons. Only the connection between neurons has modifiable synapses (weights). Recurrent synapses (connected to the same neuron) are not modifiable. We arbitrarily assigned a value of 0.7 to the weights of these synapses. The effect of recurrent connections is that the neurons' activation decays over time as it happens with cortical pyramidal neurons.

It is remarkable that Temple Fay envisioned that cybernetic principles were important to understand the central nervous system. In fact these principles are at the heart of neuronal functioning and are determinant for allowing learning in our robotic structure. These principles act in the modulation of synaptic weights (synaptic plasticity) and in the adjustment of the neurons firing interval (intrinsic plasticity).

Synaptic Plasticity. Synaptic plasticity refers to the modulation of the efficacy of information transmission between neurons, being related to the regulation of the number of ionic channels in synapses.

The first model of synaptic plasticity was postulated by Hebb and is known as the Hebb rule, that may be stated as follows: when two neurons fire together they wire together or, in other words, the synaptic strength between neurons with correlated firing tends to increase[6]. Mathematically the change in the synaptic strength (synaptic weight) between neurons i and j is calculated by the product of the output of neuron i , O_i , and the input I_j (which corresponds to the output of neuron j) multiplied by a learning constant.

$$\Delta\omega_{ij} = \varepsilon O_i I_j \quad (1)$$

Despite its importance, this formulation has a limitation: because in the equation all variables are positive, it leads synaptic weights to grow without bound. For allowing synaptic depression, alternative equations that take into account more recent biological studies have been formulated[5]. The equation that was adopted for our simulation of synaptic plasticity, due to its biological plausibility, is Grossberg's presynaptic learning rule which is as follows:

$$\Delta\omega_{ij} = \varepsilon I_j (\alpha_i - \omega_{ij}) \quad (2)$$

Where α is the activation given by the sum of synaptic contributions. In this equation, the subtractive term is equivalent to the negative feedback of a cybernetic loop. This negative feedback allows the artificial neuron to exhibit metaplasticity, a very important characteristic of biological neurons[2][8]. Metaplasticity slows down the process of weight increment or decrement, making it more difficult for the neuron to become either saturated or inactive. In the case of the equation, it is easy to understand that the increment of weight is less for big initial weights than for smaller weights.

Metaplasticity is depicted in Fig. 3, in which each curve shows the variation of weight as a function of the neuron's postsynaptic activity. The parameter that defines which curve is valid in each case is the value of the synaptic weight. According to Fig. 3, for higher values of the synaptic weight the curves are more elongated to the right.

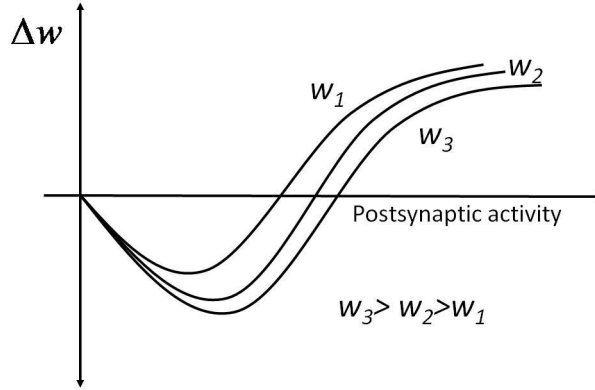


Fig. 3. Synaptic plasticity curves. The value of the initial weight determines which curve should be used for calculating weight variation. Curves with higher initial weights are more elongated to the right.

Metaplasticity regulates weight variation, down-regulating weight increment in synapses with high initial weights and up-regulating weight increment in synapses with low initial weights.

Intrinsic Plasticity. Although synaptic metaplasticity makes it difficult for synaptic weights to become either null or saturated, it does not totally preclude either of these two extreme situations. To eliminate the possibility of either weight annihilation or saturation, another important homeostatic property of real neurons should be taken into account: the so-called intrinsic plasticity.[1][9]

Intrinsic plasticity regulates the position (shift) of the neuron's activation function according to the past average level of activity in the neuron. The neuron's activation function is usually modeled as a sigmoidal function:

$$P(O) = \frac{1}{1 + e^{-25(\alpha - shift)}} \quad (3)$$

In which $P(O)$ is the output probability of the neuron and a is the activation given by the sum of synaptic contributions.

Intrinsic plasticity was modeled according to the following equation, that yields the position of the sigmoid in terms of the previous position $shift_{t-1}$ and the previous sigmoid output O_{t-1} .

$$shift_t = \frac{\xi O_{t-1} + shift_{t-1}}{1 + \xi} \quad (4)$$

Parameter ξ is a small arbitrary factor ranging in our case from 0.1 to 0.0001. This equation means that the more the neuron keeps firing, the higher will be the rightward shift of the activation function, leading to a moderation of the neurons' firing probability in the future. Conversely, if the firing probability is low, the sigmoid will move leftwards, thereby increasing the probability of the neuron's firing in the future. Parameter ξ works as a learning factor. If we want quick convergence of the shifts with little interest in accuracy, we select a higher ξ . However if we are not worried about the duration of learning and we are mainly interested in accuracy, ξ must be set to a very small value.

Competition between neurons. Another neural mechanism used in our simulation, is competition between neurons. In biological neural networks, inhibitory neurons like basket or chandelier cells are able to reduce the activity of neighboring excitatory cell so that a few excitatory ones remain active. This competition allows to find the most activated neurons inside a neural pool. In our case the most activated neuron remains active while remaining ones are inactivated.

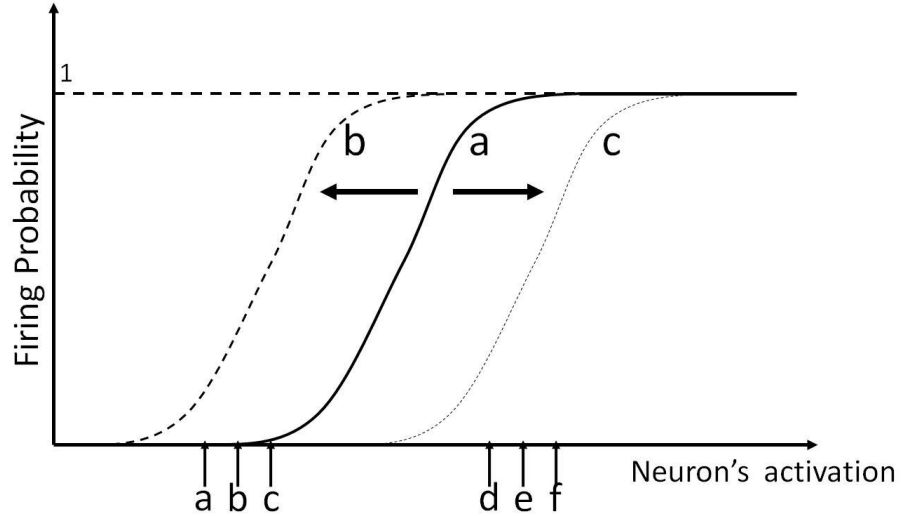


Fig. 4. According to intrinsic plasticity, low neuron activation (cases a,b,c) contributes to a leftward shift of the sigmoidal activation function. On the contrary high neuron activation (cases d,e and f) contributes to a rightward shift of this sigmoidal function.

3 Results with different kinds of stimuli

As mentioned in the introduction, the purpose of our work is to apply the inclined floor stimulation technique to a robot for understanding how this technique contributes to accelerating crawling abilities in children.

Glenn Doman suggested placing the children on a ramp in a belly-down position. The inclination of the ramp should be increased up to the point where the movements of the child make his/her body slip on the ramp. For better results, parents are suggested to shout or clap when the child successfully moves down the ramp. Black and white stripes painted on the ramp contributes to improve even more the performance of the method, also helping to stimulate children sight.

Before placing the robot on the inclined floor its four artificial neurons fire in a random way, and the wheels (controlled by these neurons) move also randomly. Once the robot is placed on the ramp, it slides a little when it moves downward (sliding does not take place when the robot moves upward). Sliding makes the sensory input of the robot being highly activated and, since sliding happens during downward movements, downward movements produce a higher sensory input. In the case of the light sensor, the higher activation takes place because the sensor identifies more transitions between white and black stripes when sliding downward than when moving upwards. In the case of the ultrasonic detector it was modified to calculate accelerations, and accelerations are also more intense when the movement of the robot is helped by gravity, i.e when it slides downwards. The only stimulus in which gravity is not relevant is sound, which in our case is produced by a maraca. The maraca is played when the robot performs a correct movement.

Our working hypothesis is that, when a random sequence of neurons is such that makes the robot goes downwards, the subsequent sensory stimuli contribute to reinforce the weights between these neurons. In this way, different chains of neurons linked by strengthened weights are formed, each chain representing a specific way the robot goes downwards. The more intense the sensory stimuli, the stronger are the connections in a chain of neurons.

To evaluate these hypotheses we counted the number of iterations (each iteration corresponding to each movement of the robot) until the robot learns to go downward under several arrangements of its environment. For example, a specific arrangement of the environment is the robot placed on the ramp (so that it experiments body acceleration) without sound and without stripes painted on the ramp (the behavior of the robot with some of these arrangements can be seen on youtube by placing the authors’ family names in the youtube searching tool). We consider that the robot has learned when it moves in the same direction during five consecutive iterations.

In Table 1 we show different arrangements of the robot environment in which different types of sensory stimuli were used. We performed 15 tests for each configuration. In each test we counted the number of iterations that it takes for the robot to learn. The mean and standard deviation of the 15 tests of each configuration are also shown in Table 1.

	Mean	Standard Deviation
Stripes	32.00	11.15
Sound	21.69	5.63
Acceleration	27.53	4.15
Stripes+Sound	19.23	3.57
Sound+Acceleration	21.15	3.44
Stripes+Acceleration	24.07	6.72
Stripes+Sound+Acceleration	19.38	2.99

Table 1. *Number of iterations that takes for the robot to learn. Each iteration corresponds to one movement of the robot.*

4 Practical implications

According to Table 1, when two stimuli are combined, the number of iterations it takes for the robot to learn is less than when a single stimulus appears alone. In the case of having three stimuli, the number of iterations that it takes for the robot to learn is less than when having two or one stimuli. This result is consistent with Doman’s methods in which parents are suggested to combine all these stimuli for improving the methods’ efficacy.

It can be also noticed that the presence of sound (the noise of a maraca when the robot performs correct movements) yields the lower number of iterations, either when the sound is alone or combined with different kinds of stimuli. This happens because the intensity of sound is always the same, regardless of how much the robot slides on the floor (which depends on the circumstances like the adjustment of the robot brakes and the slipperiness and inclination of the ramp). According to our studies, sound seems to be an important stimulus for stimulating crawling in the robot and, hypothetically in children. Clapping, shouting or even playing a musical instrument would help children to consolidate the successful motor commands that make them crawl downwards on the inclined floor. All these suggestions should be empirically tested with children.

5 Conclusion

In this work we have constructed a very simple robotic structure with three type of sensors and two independent wheel sets. A fully interconnected four-neurons neural network receive sensory inputs and controls the wheel sets. Each neuron performs a single movement of a specific wheel set. For example, when neuron 1 fires, it only produces clockwise movements in the frontal wheel set. This robotic structure was placed in an inclined floor to mimic Doman’s method for stimulating crawling abilities in children. Other stimuli like sound or alternating black and white stripes painted on the ramp are also added to the robotic environment for the purpose of mimicking the arrangement of the environment in Doman’s method.

Cybernetic principles envisioned by Doman's team are present in the way our robotic neural network works. For example, the presynaptic rule (used for synaptic weight adjustment) and our rule for modeling intrinsic plasticity are consistent with cybernetic and biological principles.

We constructed the robot using a LEGO Mindstorms NXT toolkit and programmed the neural network using the RWTH Mindstorms NXT toolbox of Matlab (which is freely available on the internet).

Despite the simplicity of the structure and the small number of neurons, the robot is capable of continuous learning, that is, it learns during the execution of its movements. In this way the robot learned what sequences of motor commands lead him downwards and what sequences failed to do this.

For simulating the different stimuli that are present in Doman's inclined floor technique, different stimuli were applied to the robotic structure. We performed 15 tests with each stimulus and calculate the average number of iterations it takes for the robot to learn. According to our expectations and consistently to Doman's empirical results, learning was quicker when more rewarding stimuli were present. Our work also suggests some improvements for the inclined floor technique that should be empirically tested.

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